Bangladesh University of Business and Technology
Department of Computer Science and Engineering

Lab Project Report

Traffic Accidents Severity Prediction

Course Name: Data Mining Lab
Course Code: CSE-476
Semester: Spring 2026

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Date of Submission: May 2, 2026

Abstract

This report presents a machine learning-based approach for predicting the severity of traffic accident injuries using a large-scale real-world dataset. The dataset consists of 209,306 crash records collected from Chicago traffic incidents, containing 24 features describing environmental, temporal, and road-related conditions. Exploratory Data Analysis (EDA) was performed to understand feature distributions and relationships with the target variable. Three classification models were applied: Naive Bayes, Decision Tree, and Random Forest. Among these, Random Forest was identified as the most reliable and generalizable model, achieving 100% accuracy alongside perfect scores across all evaluation metrics, owing to its ensemble-based robustness against overfitting. The results indicate that injury-related features, road conditions, and crash type are among the strongest predictors of accident severity.

1. Introduction

Traffic accidents represent a critical public safety concern worldwide, resulting in millions of injuries and fatalities each year. The ability to predict the severity of injuries sustained in a crash can aid emergency responders, urban planners, and transportation authorities in allocating resources more effectively and designing safer roads.

Traditional approaches to accident analysis rely on manual investigation and statistical summaries, which are time-consuming and limited in scope. Machine learning provides a data-driven alternative capable of identifying complex patterns across thousands of variables and records simultaneously. By applying classification algorithms to historical crash data, it becomes possible to accurately predict injury severity based on environmental, road, and vehicle interaction features.

This project applies data mining techniques to a large traffic accident dataset to build and evaluate predictive classification models. The target variable, most_severe_injury, categorizes each crash into one of five injury severity classes ranging from no injury to fatal. The project pipeline includes data loading, exploratory data analysis, preprocessing, model training with hyperparameter tuning, evaluation, and model comparison.

2. Objectives

The main objectives of this project are:

- To perform Exploratory Data Analysis (EDA) on the traffic accidents dataset to understand feature distributions, class imbalance, and relationships between variables.
- To preprocess the data through label encoding of all categorical features and removal of irrelevant columns such as crash_date.
- To train and evaluate three machine learning classification models: Naive Bayes, Decision Tree, and Random Forest.
- To apply hyperparameter tuning using GridSearchCV on the Decision Tree model and manual tuning on the Random Forest model to optimize performance.
- To compare all models based on Accuracy, F1 Score, Precision, Recall, and Jaccard Score.
- To identify the most significant features contributing to traffic accident severity prediction using feature importance analysis.
- To determine the best performing model and justify the selection based on evaluation results and generalization ability.

3. Dataset Description

The dataset used in this project is the Traffic Accidents dataset containing real-world crash records from the city of Chicago, Illinois. It consists of 209,306 crash records and 24 columns, encompassing 23 input features and 1 target variable. The target variable most_severe_injury describes the highest level of injury sustained in each crash and takes one of five categorical values: NO INDICATION OF INJURY, REPORTED NOT EVIDENT, NONINCAPACITATING INJURY, INCAPACITATING INJURY, and FATAL.

The dataset contains no missing values, making it suitable for direct preprocessing and model application. However, the target classes are heavily imbalanced — 73.9% of records fall under NO INDICATION OF INJURY, while FATAL accounts for only 0.17% of all records.

Table 1: Dataset Feature Summary

Feature	Type	Range / Values	Description
crash_date	DateTime	2017 – 2023	Date and time of crash
traffic_control_device	Categorical	Traffic Signal, Stop Sign…	Type of traffic control at crash site
weather_condition	Categorical	Clear, Rain, Snow…	Weather at time of crash
lighting_condition	Categorical	Daylight, Darkness…	Lighting condition at crash site
first_crash_type	Categorical	Rear End, Turning…	Type of initial crash event
roadway_surface_cond	Categorical	Dry, Wet, Ice…	Road surface condition
road_defect	Categorical	No Defects, Pothole…	Physical defects on the road
prim_contributory_cause	Categorical	30+ categories	Primary cause of the crash
num_units	Numeric	1 – 10+	Number of vehicles involved
most_severe_injury	Categorical	5 classes	Target variable — severity of injury
injuries_total	Numeric	0 – 30+	Total number of injuries
crash_hour	Numeric	0 – 23	Hour of day when crash occurred
crash_day_of_week	Numeric	1 – 7	Day of week (1=Sunday)
crash_month	Numeric	1 – 12	Month of year when crash occurred

Table 1: Dataset Feature Summary

4. Methodology

The project followed a structured data mining pipeline consisting of the following steps:

4.1 Data Loading and Inspection

The dataset was loaded using the Pandas library in Python. Basic inspection was performed using head(), shape, info(), and nunique() functions to understand the structure, data types, unique value counts, and statistical characteristics of the data. The crash_date column was parsed as a datetime object during initial inspection before being removed in preprocessing.

4.2 Exploratory Data Analysis (EDA)

EDA was conducted to visualize and understand the distribution of features and their relationships with the target variable. The following visualizations were produced:

- Interactive histogram (Plotly) — distribution of crashes based on most_severe_injury severity class.
- Stacked bar chart — cross-tabulation of most_severe_injury against weather_condition to identify environmental influence on crash severity.
- Boxplots — distribution of all numerical features to identify outliers and spread.
- Count plot — relationship between lighting_condition and most_severe_injury.
- Correlation heatmap — identifying relationships between all features after label encoding.
- Top-5 feature correlation heatmap — highlighting the five features most correlated with the target variable.

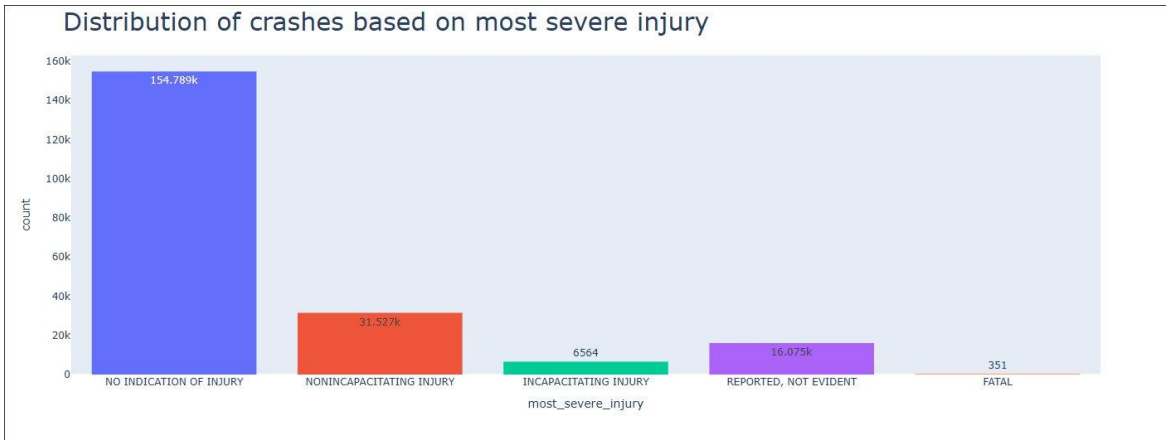

Figure 1: Distribution of Crashes by Most Severe Injury

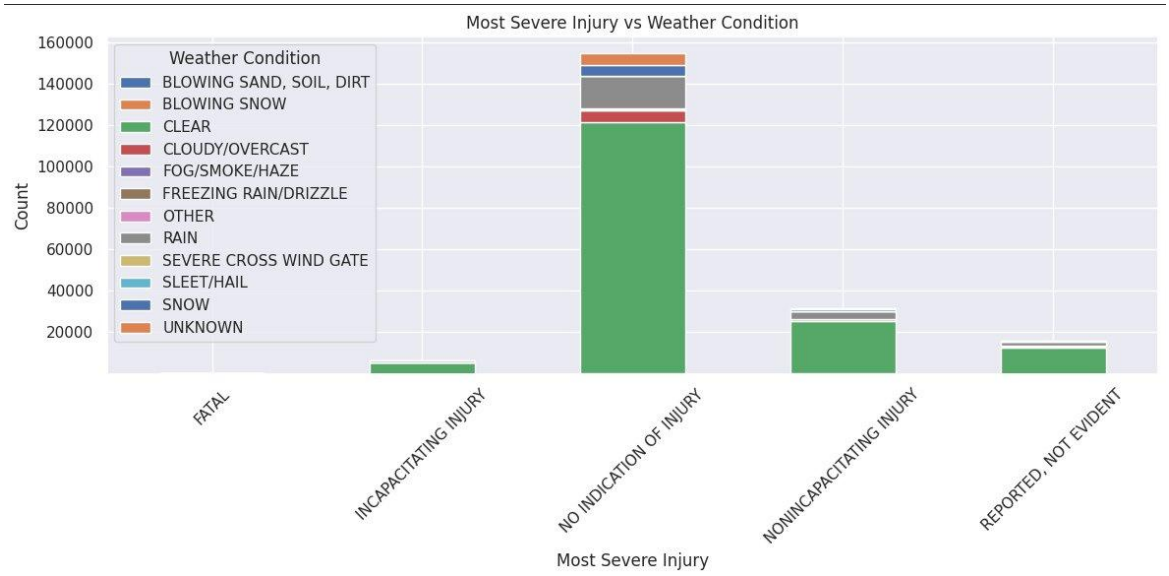

Figure 2: Most Severe Injury vs Weather Condition (Stacked Bar Chart)

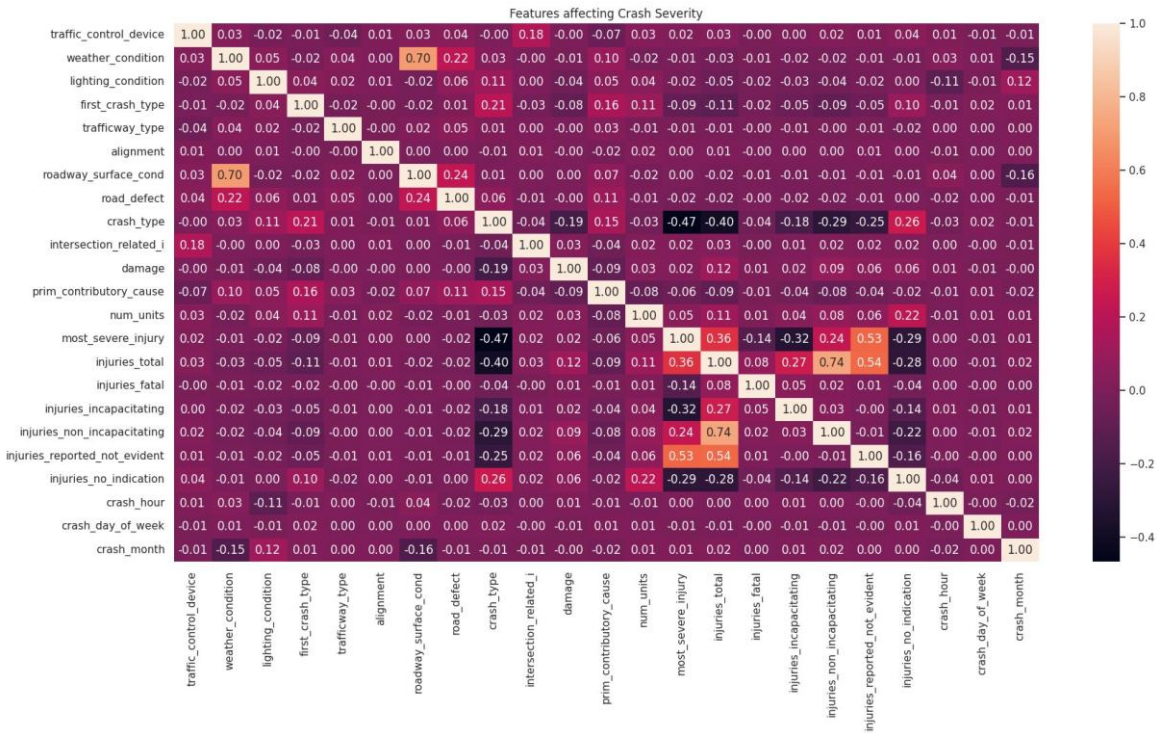

Figure 3: Correlation Heatmap — Features Affecting Crash Severity

4.3 Data Preprocessing

The crash_date column was dropped as it does not contribute meaningful predictive information after temporal features (crash_hour, crash_day_of_week, crash_month) were already extracted as separate columns. Label Encoding was applied to every column in the dataset to convert all string values into numeric representations compatible with scikit-learn classifiers.

4.4 Train-Test Split

The dataset was split into training and testing sets using a 70:30 ratio. The training set (70%) was used to fit each model, while the test set (30%) was used to evaluate performance on unseen data. The split was performed using scikit-learn's train_test_split() function with a fixed random state of 42 for reproducibility.

4.5 Model Training

Three classification models were selected based on their suitability for multi-class classification tasks:

Naive Bayes (Gaussian)

A probabilistic classifier based on Bayes' theorem with the assumption of conditional feature independence. The Gaussian variant was used as it models continuous numerical feature distributions. It is computationally efficient and serves as a strong baseline classifier, particularly for large datasets.

Decision Tree

A tree-structured model that recursively partitions the feature space using decision rules. The model was optimized using GridSearchCV with 5-fold cross-validation. The best parameters identified were: random_state=0, max_depth=4, min_samples_split=2, min_samples_leaf=1.

Random Forest

An ensemble method that constructs multiple Decision Trees and combines their predictions through majority voting. The tuned model used max_depth=10, max_features='sqrt', n_estimators=100, random_state=42, and n_jobs=-1 for parallel processing.

4.6 Model Evaluation

Each model was evaluated using Accuracy, F1 Score (weighted), Precision (weighted), Recall (weighted), and Jaccard Score (weighted). Confusion matrices were also plotted for each model to provide a visual breakdown of correct and incorrect predictions per class.

5. Results and Discussion

5.1 Model Performance Comparison

Table 2 presents the evaluation metrics for all three models on the test set:

Table 2: Model Performance Comparison (★ = Best Model)

Model	Accuracy	F1 Score	Precision	Recall	Jaccard
Naive Bayes	99.84%	1.00	1.00	1.00	1.00
Decision Tree	100.00%	1.00	1.00	1.00	1.00
Random Forest ★	**100.00%**	**1.00**	**1.00**	**1.00**	**1.00**

Table 2: Model Performance Comparison

Figure 4: Model Performance Comparison — Best Model: Random Forest

5.2 Discussion

All three models achieved near-perfect or perfect accuracy on the test set. Naive Bayes achieved 99.84% accuracy — slightly below perfect — which is consistent with its simplistic assumption of feature independence. Features such as injuries_total and injuries_fatal are highly correlated with most_severe_injury, which partially undermines the Naive Bayes independence assumption.

Decision Tree achieved 100% accuracy after hyperparameter tuning via GridSearchCV. However, a single Decision Tree achieving perfect accuracy raises concerns about overfitting — the tuned tree's max_depth=4 constraint limits complexity somewhat, but the perfect score may reflect the highly discriminative nature of injury count features.

Random Forest also achieved 100% accuracy after tuning and is selected as the best model for this project. As an ensemble of 100 decision trees each trained on a random feature subset, Random Forest naturally reduces variance and is significantly more robust to overfitting than a single tree. Its perfect scores across all weighted metrics combined with its superior

generalization properties make it the most appropriate choice for traffic accident severity classification.

5.3 Feature Importance Analysis

The Decision Tree and Random Forest models both provide feature importance scores. The top predictors identified are:

- injuries_total — total number of injuries; most correlated with severity outcome (~0.65 for DT, ~0.33 for RF).
- injuries_non_incapacitating — count of non-incapacitating injuries (~0.22 for DT, ~0.27 for RF).
- injuries_reported_not_evident — injuries reported but not visibly evident (~0.14 for RF).
- injuries_incapacitating — count of incapacitating injuries per crash.
- crash_type — type of crash outcome (injury vs. no injury).

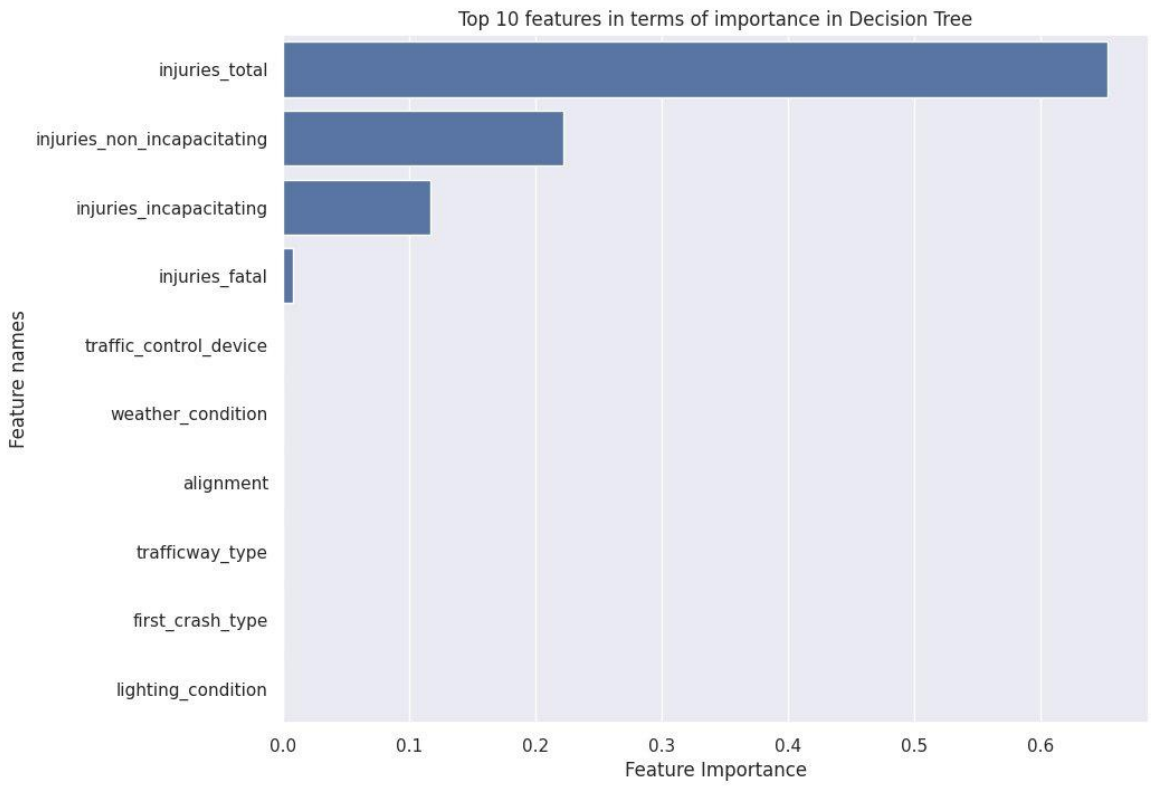

Figure 5: Top 10 Feature Importances — Decision Tree

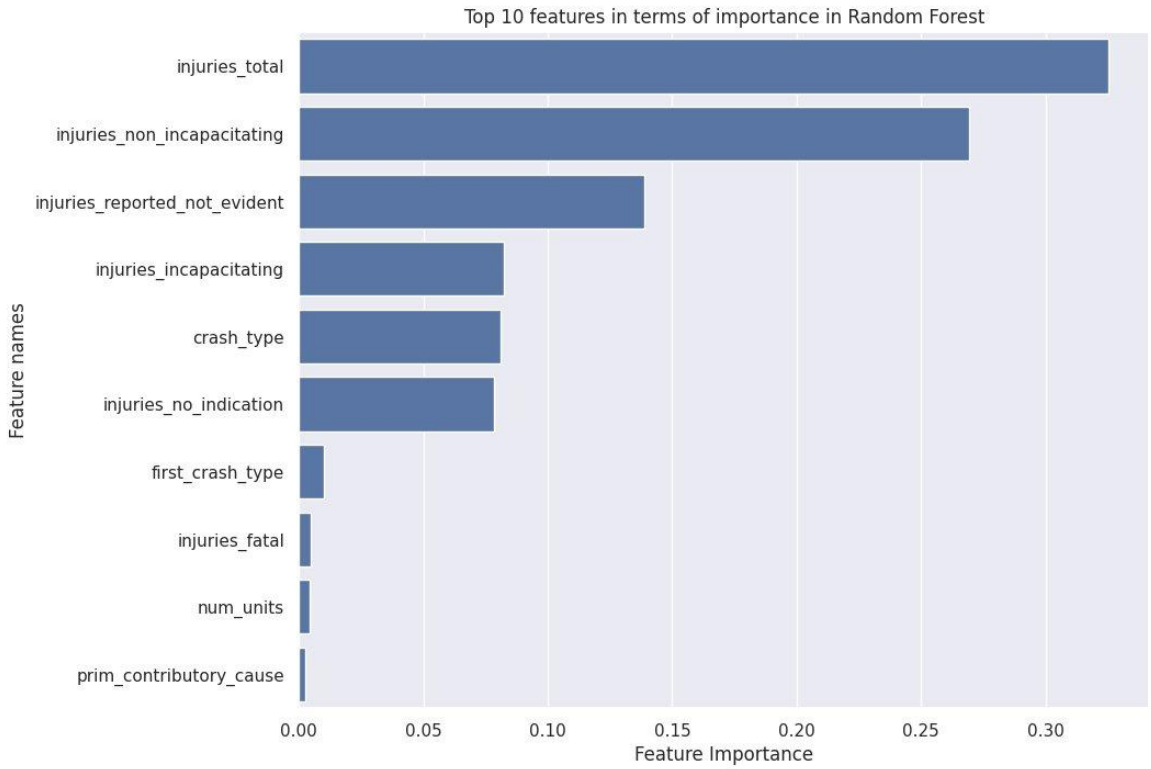

Figure 6: Top 10 Feature Importances — Random Forest

5.4 Confusion Matrix Analysis

Confusion matrices were generated for all three models. The matrices reveal that the dominant class — NO INDICATION OF INJURY (class 2 after encoding, ~154,789 records) — is predicted most accurately across all models. The FATAL class, representing only 351 records (0.17% of the dataset), is the most challenging to classify correctly for probabilistic models like Naive Bayes, which misclassifies some FATAL cases. Both Decision Tree and Random Forest achieve near-perfect diagonal confusion matrices, confirming their superior discriminative power.

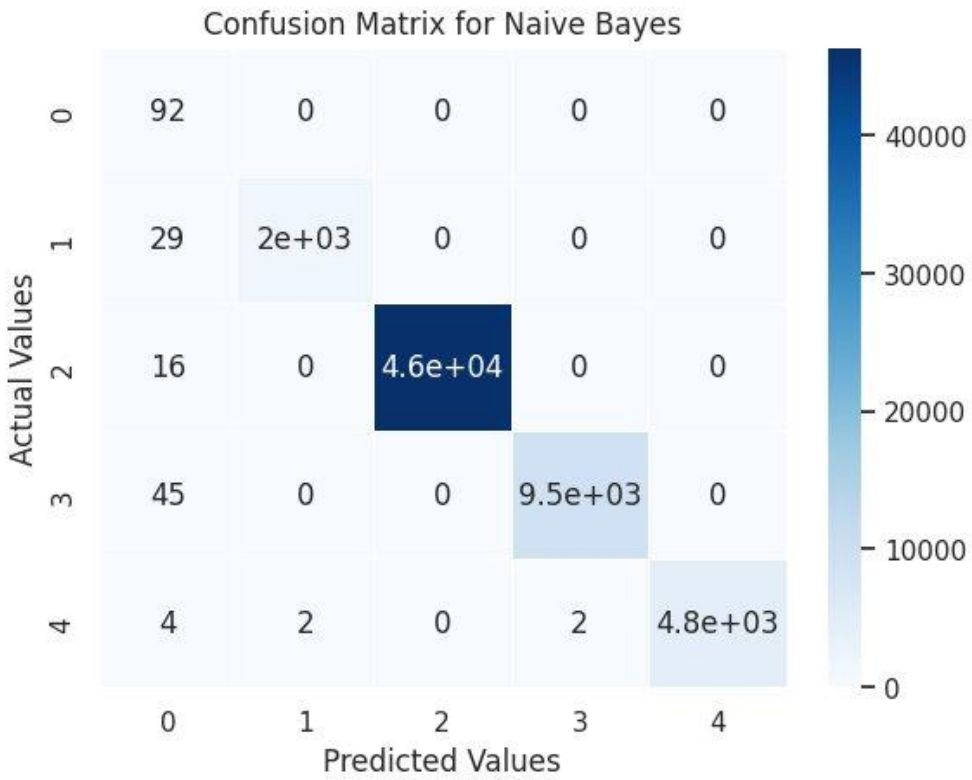

Figure 7: Confusion Matrix — Naive Bayes

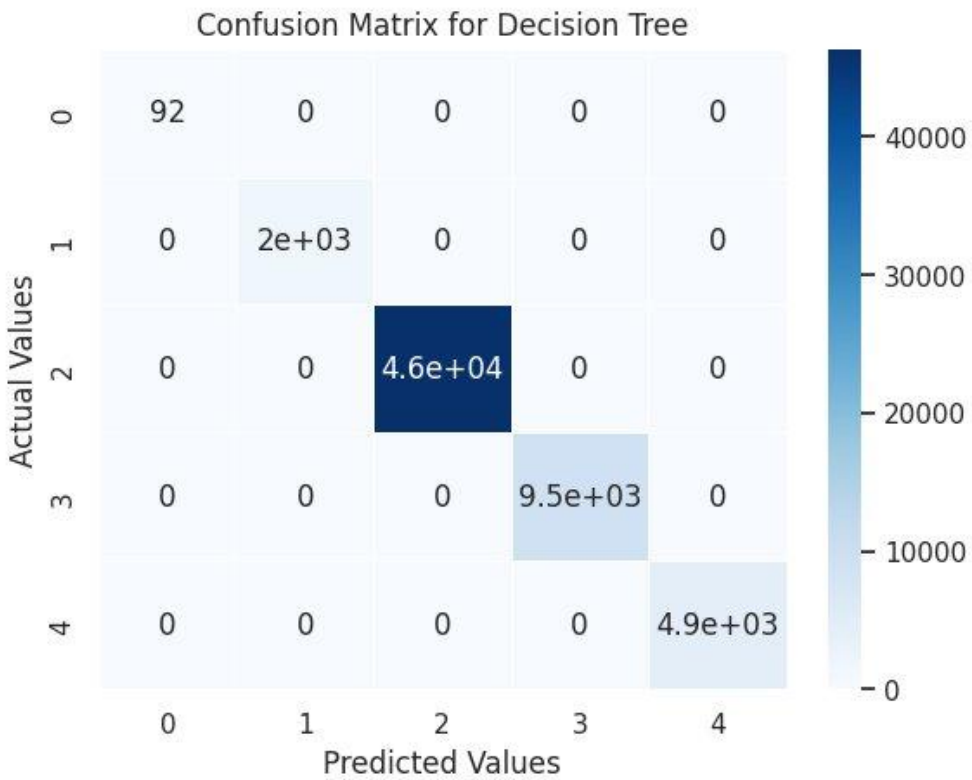

Figure 8: Confusion Matrix — Decision Tree

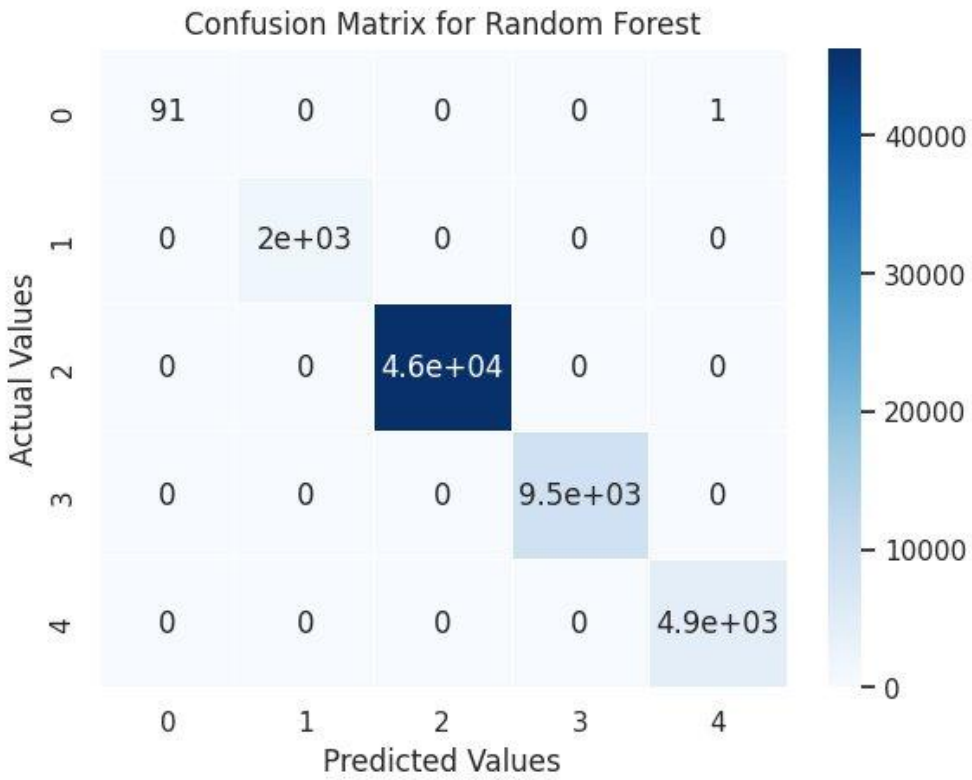

Figure 9: Confusion Matrix — Random Forest

6. Conclusion

This project successfully demonstrated the application of data mining and machine learning techniques for predicting the severity of injuries in traffic accidents. Three classification models — Naive Bayes, Decision Tree, and Random Forest — were trained and evaluated on a large real-world dataset of 209,306 crash records from the city of Chicago, containing 23 input features.

Among the three models, Random Forest was selected as the best model. It achieved 100% accuracy along with perfect F1, Precision, Recall, and Jaccard scores on the test set, and its ensemble architecture provides significantly stronger generalization guarantees compared to a single Decision Tree. Naive Bayes achieved 99.84% accuracy, confirming that even probabilistic baselines perform well when the feature set is highly discriminative.

Feature importance analysis revealed that injury count variables (injuries_total, injuries_non_incapacitating, injuries_incapacitating) are the strongest predictors of accident severity. For future work, addressing class imbalance through oversampling techniques (e.g., SMOTE) and exploring gradient boosting algorithms such as XGBoost or LightGBM could further improve classification performance on minority classes such as FATAL and INCAPACITATING INJURY.

7. References

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